

100 Days of Linux - Day 042

Title: Compressing Files with gzip and gunzip

Today's Goal

Learn how to compress and decompress files using **gzip** and **gunzip**, and understand how they work with TAR archives.

Why This Matters

Compression reduces file size, making backups faster and file transfers more efficient. Data engineers frequently compress logs, datasets, and backups.

Commands of the Day: gzip & gunzip

Common Syntax:

```
gzip filename
gunzip filename.gz
gzip -k filename (keep original file)
gzip -l filename.gz (view compression information)
```

Common TAR Combination:

```
tar -czvf backup.tar.gz project/
tar -xzvf backup.tar.gz
```

Options:

- k = Keep the original file
- l = List compression information
- z = Use gzip with tar

Beginner Lesson

Unlike tar, gzip compresses files. By default, gzip replaces the original file with a compressed .gz file. When you need to archive an entire folder, tar and gzip are commonly used together to create a .tar.gz archive.

Practice Questions

1. Create a file named report.txt and compress it using gzip.
2. Decompress report.txt.gz using gunzip and verify the original file is restored.
3. Create another file named logs.txt and compress it while keeping the original file.
4. Create a folder named demo_project containing two files. Create a compressed archive named demo_project.tar.gz using tar and gzip together.
5. Display your current working directory and list all .gz files.

Hints

Remember: gzip works on files, while tar -czvf is commonly used for folders.

Mini Challenge

Create a data_backup folder with CSV files and logs. Build a

data_backup.tar.gz archive, extract it into a restore folder, and verify every file.

Common Mistakes

Trying to compress a directory directly with gzip. Use tar with the -z option for folders.

Did You Know?

Many Linux software packages and source code releases are distributed as .tar.gz archives.

Pro Tip

Use gzip -k when you want to keep both the original file and the compressed copy.

Answer Key (Instructor Only)

Q1: gzip report.txt

Q2: gunzip report.txt.gz

Q3: gzip -k logs.txt

Q4: tar -czvf demo_project.tar.gz demo_project/

Q5: pwd && ls *.gz

Homework

Practice creating and extracting several .tar.gz archives until you can do it without referring to your notes.

Next Day Preview

Tomorrow you'll learn zip and unzip to create archives that are commonly shared across Windows, macOS, and Linux.